

TP LSQL pour DA2I- L2- EMIT FIANARANTSOA (2025- 2026)

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A : QCM

1. De quoi SQL est-il l'acronyme ?
Structured Query Language
2. Quel mot-clé SQL est utilisé pour rechercher des données dans une base de données ?
SELECT
3. Quel mot-clé SQL est utilisé pour modifier des données dans une base de données ?
UPDATE
4. Quel mot-clé SQL est utilisé pour ajouter des données dans une base de données ?
INSERT
5. Quel mot-clé SQL est utilisé pour supprimer des données dans une base de données ?
DELETE
6. Quel(s) mot-clé(s) SQL est utilisé pour trier les résultats d'une requête ?
ORDER BY

B : TRAVAUX PRATIQUES

TP1 : LES JEUX OLYMPIQUES

1. Explication des requêtes
 - a. Rechercher et retourne toutes les données dans la table Medals qui remplissent ces conditions où l'année est en 2008 (Year=2008) et aussi le pays est La France (Country= «France»).
 - b. Rechercher et retourne les attributs City, Year, Athlete pour toutes les données dans la table Medals qui remplissent les conditions où le médaille est doré et l'évènement est 110m hurdles (Medal= «Gold» AND Event= «110 hurdles»).

```
mysql> SELECT City, Year, Athlete
-> FROM Medals
-> WHERE Medal="Gold" AND Event="110m hurdles";
```

City	Year	Athlete
Montreal	1976	DRUT, Guy
Moscow	1980	MUNKELT, Thomas
Los Angeles	1984	KINGDOM, Roger
Seoul	1988	KINGDOM, Roger
Barcelona	1992	MCKOY, Mark
Atlanta	1996	JOHNSON, Allen
Sydney	2000	GARCIA, Anier
Athens	2004	LIU, Xiang
Beijing	2008	ROBLES, Dayron

```
9 rows in set (0.247 sec)
```

- c. Rechercher et retourne les attributs Athlete, Event pour toutes les données dans la table Medals qui remplissent les conditions où le médaille est doré et le pays qui a gagné la médaille est La France et enfin l'année de l'obtention de la médaille est l'ans 2000 et toutes données sont ordonnées en **Ordre croissante Alphabétiquement (A – Z)**.

```
mysql> SELECT Athlete, Event
-> FROM Medals
-> WHERE Medal="Gold" AND Country="France" AND Year=2000
-> ORDER BY Athlete ASC;
```

Athlete	Event
ANDRIEUX, Michel	coxless pair (2-)
ASLOUM, Brahim	- 48kg (light-flyweight)
BALLANGER, Felicia	500m time trial
BALLANGER, Felicia	sprint
BETTE, Jean-Christophe	lightweight coxless four (4-)
DORFMANN, Xavier	lightweight coxless four (4-)
DOUILLET, David	+ 100kg (heavyweight)
DUMOULIN, Franck	10m air pistol (60 shots)
ESTANGUET, Tony	C-1 (canoe single)
FERRARI, Jean-Noel	foil team
GANE, Laurent	Olympic Sprint
GUYART, Brice	foil team
HOCDE, Yves	lightweight coxless four (4-)
LHOTELLIER, Patrice	foil team
MARTINEZ, Miguel	cross-country
PLUMENAIL, Lionel	foil team
PORCHIER, Laurent	lightweight coxless four (4-)
ROLLAND, Jean Christophe	coxless pair (2-)
ROUSSEAU, Florian	Olympic Sprint
ROUSSEAU, Florian	Keirin
TOURNANT, Arnaud	Olympic Sprint
VANDENHENDE, Severine	57 - 63kg (half-middleweight)

```
22 rows in set (0.180 sec)
```

- d. Rechercher et retourne les attributs Country (pays) et élimine les ligne identiques donc si France a passé, elle ne reviendra plus dans le résultat, et pour toutes les données dans la table Medals qui remplissent les conditions où le médaille est doré et l'année est en 1984 en **Ordre décroissante alphabétiquement (Z – A)**.

```
mysql> SELECT DISTINCT Country
-> FROM Medals
-> WHERE Medal="Gold" AND YEAR="1984" ORDER BY Country DESC;
```

Country
Yugoslavia
West Germany
United States
United Kingdom
Sweden
Spain
Romania
Portugal
Pakistan
New Zealand
Netherlands
Morocco
Mexico
Korea, South
Kenya
Japan
Italy
France
Finland
China
Canada
Brazil
Belgium
Austria
Australia

25 rows in set (0.278 sec)

- e. Rechercher et retourne les attributs Athlete, City, Year, Medal, Country pour toutes les données dans la table Medals que seulement avec **les pays dont le nom se termine avec « ..nia»** peut import s'il n'y a aucun ou plusieurs caractères devant « ..nia» du nom du pays donc la longueur du caractère >= 3 (Country LIKE «%nia»).

```
mysql> SELECT Athlete, City, Year, Medal, Country
-> FROM Medals
-> WHERE Country LIKE '%nia';
```

Athlete	City	Year	Medal	Country
MEGELEA, Gheorghe	Montreal	1976	Bronze	Romania
DAFINOIU, Costica	Montreal	1976	Bronze	Romania
SIMON, Mircea	Montreal	1976	Silver	Romania
ZILBERMAN, Victor	Montreal	1976	Bronze	Romania
CUTOV, Simion	Montreal	1976	Silver	Romania
NASTAC, Alexandru	Montreal	1976	Bronze	Romania
DIBA, Vasile	Montreal	1976	Bronze	Romania
DIBA, Vasile	Montreal	1976	Gold	Romania
SERHEI, Larion	Montreal	1976	Bronze	Romania
SIMIONOV, Gheorghe	Montreal	1976	Silver	Romania
MALIHIN, Policarp	Montreal	1976	Bronze	Romania
DANIELOV, Gheorghe	Montreal	1976	Silver	Romania
NILCA, Alexandru	Montreal	1976	Bronze	Romania
POP, Ioan	Montreal	1976	Bronze	Romania
MARIN, Corneliu	Montreal	1976	Bronze	Romania
IRIMICIUC, Daniel	Montreal	1976	Bronze	Romania
MUSTATA, Marin	Montreal	1976	Bronze	Romania

2. Requête SQL

- a.

```
mysql> SELECT DISTINCT Athlete
-> FROM Medals
-> WHERE Medal="Gold" AND Country='France' AND Year=1984;
```

Athlete
QUINON, Pierre
LAMOUR, Jean-Francois
BOISSE, Philippe
BIBARD, Michel
RUST, Albert
GARANDE, Patrice
ZANON, Jean Louis
BRISSON, Francois
CUBAYNES, Patrick
BENSOUSSAN, Michel
THOUVENEL, Jean Christophe
LACOMBE, Guy
LEMOULT, Jean-Claude
AYACHE, William
BIJOTAT, Dominique
SENAC, Didier
JEANNOL, Philippe
TOURE, Jose
XUEREB, Daniel
ROHR, Jean Philippe
HEBERLE, Philippe

```
21 rows in set (0.157 sec)
```

b.

```
mysql> SELECT DISTINCT Athlete, Gender, Sport, Year
-> FROM Medals
-> WHERE Gender='Women' AND Sport='Marathon'
-> ORDER BY Year ASC;
Empty set (0.247 sec)
```

c.

```
mysql> SELECT DISTINCT Athlete, Medal, Discipline
-> FROM Medals
-> WHERE Athlete LIKE '%Adrian%';
```

Athlete	Medal	Discipline
DANTLEY, Adrian Delano	Gold	Basketball
COSMA, Adrian	Silver	Handball
PARKER, Adrian Philip	Gold	Modern Pentath.
COSMA, Adrian	Bronze	Handball
SIMION, Adrian	Bronze	Handball
ELLISON, Adrian Charles	Gold	Rowing
BAZON, Adriana	Silver	Rowing
MOORHOUSE, Adrian David	Gold	Swimming
DUNAVSKA, Adriana	Silver	Rhythmic G.
BECKER, Nick Adrian	Bronze	Volleyball
SANTOS, Adriana Aparecida	Silver	Basketball
SAMUEL RAMOS, Adriana	Silver	Beach volley.
SANTOS, Adriana Aparecida	Bronze	Basketball
PINTO, Adriana	Bronze	Basketball
BEHAR, Adriana	Silver	Beach volley.
SAMUEL RAMOS, Adriana	Bronze	Beach volley.
BURNSIDE, Adrian	Silver	Baseball
CARMONA, Adriana	Bronze	Taekwondo
ADRIAN, Nathan	Gold	Swimming

```
19 rows in set (0.218 sec)
```

d.

```
mysql> SELECT Athlete, Country, Medal, Year
-> FROM Medals
-> WHERE Discipline='Swimming' AND Athlete LIKE '%PHELPS%Michael%' AND Year='2004';
```

Athlete	Country	Medal	Year
PHELPS, Michael	United States	Gold	2004
PHELPS, Michael	United States	Gold	2004
PHELPS, Michael	United States	Gold	2004
PHELPS, Michael	United States	Gold	2004
PHELPS, Michael	United States	Bronze	2004
PHELPS, Michael	United States	Gold	2004
PHELPS, Michael	United States	Gold	2004
PHELPS, Michael	United States	Bronze	2004

```
8 rows in set (0.165 sec)
```

e.

```
mysql> SELECT COUNT(*) AS "Nb medaille Australien"
-> FROM Medals
-> WHERE (Medal='Gold' OR Medal='Silver') AND Country='Australia';
```

Nb medaille Australien
486

```
1 row in set (0.201 sec)
```

3.

```
mysql> SELECT Discipline, COUNT(*)
-> FROM Medals
-> WHERE Country='France' AND Medal='Gold'
-> GROUP BY Discipline
-> ORDER BY COUNT(*) DESC;
```

Discipline	COUNT(*)
Fencing	48
Football	17
Handball	15
Cycling Track	13
Judo	10
Sailing	8
Rowing	8
Athletics	6
Jumping	5
Canoe / Kayak S	5
Eventing	5
Shooting	3
Mountain Bike	3
Swimming	2
Archery	1
Cycling Road	1
Boxing	1
Artistic G.	1
BMX	1
Wrestling Gre-R	1

20 rows in set (0.257 sec)

TP2 : LES PRIX NOBEL

1. Prise de connaissance des attributs
2. Ecriture requêtes SQL
3.
 - a.

```
mysql> SELECT DISTINCT Category
-> FROM nobel;
```

Category
Chemistry
Literature
Medicine
Peace
Physics
Economics

6 rows in set (0.057 sec)

- b.

```
mysql> SELECT DISTINCT LaureateID, FullName, Year
-> FROM nobel
-> WHERE (BirthCountry='France' OR OrganizationCountry='France')
-> ORDER BY FullName ASC;
```

LaureateID	FullName	Year
NULL	Alain Aspect	2022
814	Albert Fert	2007
306	Alexis Carrel	1912
87	Alfred Kastler	1966
174	Alfred Werner	1913
590	Anatole France	1921
360	Andr�� Fr��d��ric Cournand	1956
381	Andr�� Lwoff	1965
618	Andr�� Paul Guillaume Gide	1947
NULL	Anne L'Huillier	2023
NULL	Annie Ernaux	2022
4	Antoine Henri Becquerel	1903
490	Aristide Briand	1926
25	Charles Edouard Guillaume	1920
318	Charles Jules Henri Nicolle	1928
300	Charles Louis Alphonse Laveran	1907
307	Charles Robert Richet	1913
153	Claude Cohen-Tannoudji	1997

c.

```
mysql> SELECT OrganizationName, Year
-> FROM nobel
-> WHERE OrganizationName='Comit   international de la Croix Rouge';
Empty set (0.029 sec)
```

d.

```
mysql> SELECT COUNT(*) AS "Nombre de laureate"
-> FROM nobel
-> WHERE sex='Female';
```

Nombre de laureate
69

1 row in set (0.029 sec)

e.

```
mysql> SELECT FullName, Sex, Category, Year
-> FROM nobel
-> WHERE Sex='Female' AND BirthCountry='France' AND Year >= 2005;
```

FullName	Sex	Category	Year
Fran��oise Barr��-Sinoussi	Female	Medicine	2008
Emmanuelle Charpentier	Female	Chemistry	2020
Annie Ernaux	Female	Literature	2022
Esther Duflot	Female	Economics	2019
Anne L'Huillier	Female	Physics	2023

5 rows in set (0.035 sec)

f.

```
mysql> SELECT DISTINCT FullName, (Year - CAST(BirthDate AS SIGNED)) AS "Age", DeathDate
-> FROM nobel
-> WHERE DeathDate IS NULL
-> ORDER BY Age DESC;
```

FullName	Age	DeathDate
Isamu Akasaki	85	NULL
William C. Campbell	85	NULL
Youyou Tu	85	NULL
Peter W. Higgs	84	NULL
Martin Karplus	83	NULL
Alice Munro	82	NULL
David J. Thouless	82	NULL
François Englert	81	NULL
Akira Suzuki	80	NULL
Osamu Shimomura	80	NULL

g.

```
mysql> SELECT DISTINCT FullName
-> FROM nobel
-> WHERE FullName LIKE '%curie%';
```

FullName
Pierre Curie
Marie Curie, née Skłodowska
Irène Joliot-Curie

3 rows in set (0.056 sec)

4. a.

```
mysql> INSERT INTO nobel(Year, Category, LaureateType, FullName, BirthCountry, Sex)
-> VALUES(2026, 'Chemistry', 'Individual', 'Anthonio Adrian MORA', 'Madagascar', 'Male');
Query OK, 1 row affected (0.262 sec)
```

b.

```
mysql> UPDATE nobel
-> SET Category='NSI'
-> WHERE FullName LIKE 'Anthonio Adrian MORA';
Query OK, 1 row affected (0.178 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT FullName, Category, Sex
-> FROM nobel
-> WHERE FullName LIKE 'Anthonio Adrian MORA';
```

FullName	Category	Sex
Anthonio Adrian MORA	NSI	Male

1 row in set (0.046 sec)

c.

```
mysql> DELETE
-> FROM nobel
-> WHERE FullName LIKE 'Antonio Adrian MORA';
Query OK, 1 row affected (0.173 sec)

mysql> SELECT FullName, Category, Sex
-> FROM nobel
-> WHERE FullName LIKE 'Antonio Adrian MORA';
Empty set (0.032 sec)
```

TP3 : LES JOUEURS DE BASKET EVOLUANT EN NBA

1.

```
mysql> SELECT * FROM joueurs_nba;
```

Name	Team	Number	Position	Age	Height	Weight
Steven Adams	Houston Rockets	12	C	32	211	120
Bam Adebayo	Miami Heat	13	C	28	206	116
Ochai Agbaji	Toronto Raptors	30	SF	25	196	98
Santi Aldama	Memphis Grizzlies	7	C	24	213	98
Trey Alexander	New Orleans Pelicans	23	SG	23	193	84
Nickeil Alexander-Walker	Atlanta Hawks	7	SG	27	196	93
Grayson Allen	Phoenix Suns	8	SG	29	193	90
Jarrett Allen	Cleveland Cavaliers	31	C	27	206	110

2. Cette requête retourne l'âge moyen des joueurs dans l'équipe Los Angeles Lakers grâce à la fonction AVG().

3. Ecriture requête SQL

a.

```
mysql> SELECT MAX(Height) AS "Taille Maximal"
-> FROM joueurs_nba
-> WHERE Team='Chicago Bulls';
```

Taille Maximal
211

1 row in set (0.113 sec)

b.

```
mysql> SELECT MIN(Weight) AS 'Poids le plus faible'
-> FROM joueurs_nba
-> WHERE Position='SG';
```

Poids le plus faible
75

1 row in set (0.025 sec)

c.

```
mysql> SELECT * FROM joueurs_nba WHERE Name='Victor Wembanyama';
```

Name	Team	Number	Position	Age	Height	Weight
Victor Wembanyama	San Antonio Spurs	1	F	21	221	107

```
1 row in set (0.026 sec)
```

```
mysql> SELECT Name, Age
-> FROM joueurs_nba
-> WHERE Height>210 AND Age=(SELECT MIN(Age) FROM joueurs_nba WHERE Height>210);
```

Name	Age
Khaman Maluach	19

```
1 row in set (0.028 sec)
```

d.

```
mysql> INSERT INTO joueurs_nba
-> VALUES('Anthonio Adrian MORA', 'Boston Celtics', 99, 'PK', 19, 185, 70);
Query OK, 1 row affected (0.173 sec)
```

```
mysql> SELECT * FROM joueurs_nba
-> WHERE Team='Boston Celtics';
```

Name	Team	Number	Position	Age	Height	Weight
Chris Boucher	Boston Celtics	25	PF	32	206	91
Jaylen Brown	Boston Celtics	7	SF	28	198	101
Luka Garza	Boston Celtics	52	C	26	208	110
Hugo Gonzalez	Boston Celtics	28	F	19	198	91
Hayden Gray	Boston Celtics	NULL	G	22	193	86
Sam Hauser	Boston Celtics	30	SF	27	201	99
R.J. Luis	Boston Celtics	12	F	24	201	98
Josh Minott	Boston Celtics	8	F	22	203	93
Payton Pritchard	Boston Celtics	11	PG	27	185	89
Neemias Queta	Boston Celtics	88	C	26	213	113
Baylor Scheierman	Boston Celtics	55	PG	24	198	93
Max Shulga	Boston Celtics	29	G	24	196	95
Anfernee Simons	Boston Celtics	4	SG	26	191	91
Jayson Tatum	Boston Celtics	0	SF	27	203	95
Xavier Tillman, Sr.	Boston Celtics	26	PF	26	201	111
Jordan Walsh	Boston Celtics	27	F	21	198	93
Derrick White	Boston Celtics	9	G	31	193	86
Amari Williams	Boston Celtics	77	PF	24	211	114
Anthonio Adrian MORA	Boston Celtics	99	PK	19	185	70

```
19 rows in set (0.027 sec)
```

e.

```
mysql> UPDATE joueurs_nba
-> SET Team='San Antonio Spurs', Number=75
-> WHERE Name='Nicolas Batum';
Query OK, 1 row affected (0.209 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT *
-> FROM joueurs_nba
-> WHERE Name='Nicolas Batum';
```

Name	Team	Number	Position	Age	Height	Weight
Nicolas Batum	San Antonio Spurs	75	F	36	203	104

```
1 row in set (0.030 sec)
```

f.

```
mysql> SELECT *  
      -> FROM joueurs_nba  
      -> WHERE Team='Gotham City';
```

Name	Team	Number	Position	Age	Height	Weight
Batman	Gotham City	77	PF	25	182	78
Robin	Gotham City	21	SF	19	172	69
Wonder Woman	Gotham City	6	PG	22	174	63

```
3 rows in set (0.029 sec)
```

```
mysql> DELETE FROM joueurs_nba  
      -> WHERE Team='Gotham City';
```

```
Query OK, 3 rows affected (0.144 sec)
```